> *AI-Generated* => Claude[sonnet-5], see:
>
> - https://gitlab.com/ub-dems/ds-labs/dve-sample-py/-/blob/main/notes/howtos/Python-Q6-Data-Analysis-Notebook.pdf
> - https://claude.ai/share/93ca5435-8dc0-4e8c-be05-125a23442f75

> **ONLINE** | project | source | notebook |

Climate Crisis - Europe Summer Temperature/Rainfall Trends

Overview

Objective and structure

This notebook provides:

1. **Climate analysis** — analyse **June/July** summer temperature (TG) and rainfall (RR) for European cities from all available historic data
   - forecast each country's series 10 years forward with a state-space model that reports different confidence interval (CI),
   - compare "past" (2005-2015 baseline) against "current" city-level values with pair of synchronised geographic plots and a synthetic "delta" plot with differences.

In addition:

2. **Engine benchmark** — evaluate two alternative solutions for local, file-backed analytical processing:

   - modern Rust-based `duckdb` + `polars` pipeline
   - traditional `sqlite` + `pandas` pipeline These pipelines are tested on the same data and the same three operations: ingestion, a database-side SQL join, and an in-memory DataFrame join. See: **Appendix-A** for benchmark results.

The notebook is organised as:

- **Part 0** — this documentation, references, and package initialization.
- **Part 1** — online dataset retrieval and local caching.
- **Part 2** — local database loading.
- **Part 3** — data analysis: country-level time series with forecast, and city-level past/current tuples.
- **Part 4** — data visualization:
  - **4.a**: country time series with forecasts
  - **4.b**: geographic heatmaps for Europe region
  - **4.c**: data distribution and outlier visualization
  - **4.d**: Temerature/Rainfall measurements by cities

In appendix:

- **Appendix A** — sqlite/pandas vs duckdb/polars database loading benchmarks.

Dataset sources

- **Climate data**: European Climate Assessment & Dataset (ECA&D), daily blended series, elements TG (mean temperature, 0.1 °C) and RR (daily rainfall, 0.1 mm). Source: https://www.ecad.eu/dailydata/predefinedseries.php Licence: ECA&D data policy — free for non-commercial research and education, provided ECA&D and the data providers are acknowledged (https://www.ecad.eu/download/millennium/data_policy.php). If the blended series bulk archive cannot be retrieved or exceeds the 100 MB compressed-size budget, the notebook falls back to the Open-Meteo Historical Weather API (https://open-meteo.com/en/docs/historical-weather-api), which is CC-BY-4.0 licensed and requires no API key. The fallback trigger and the actually-used source are both logged in Part 1.
- **Geographic enrichment**: GeoNames `cities500` dataset (https://download.geonames.org/export/dump/), CC-BY-4.0 licensed. Used to attach latitude, longitude, and ISO country code to each city and to restrict the analysis to European countries.

Both sources are free, bulk-downloadable, and well under the 100 MB compressed-size limit (ECA&D blended TG/RR subsets are a few MB per element when restricted to Europe; `cities500.zip` is ~11 MB compressed). Any substitution actually made at run time is recorded as a printed note in Part 1, not silently assumed here.

Methods

- **Forecast**: an `UnobservedComponents` local-level-plus-trend state-space model (`statsmodels.tsa.statespace.structural`) is fit per country per variable on the June/July yearly mean series, then used to extrapolate 10 years forward with a 50%, 80%, 95% confidence intervals. No seasonal term is included because the series is already collapsed to one observation per year (the June/July mean), so there is no within-series seasonal cycle left to model. The rationale is expanded in Part 3.
- **Geographic comparison**: city-level June/July means are aggregated into a 2005-2010 "past" baseline and a current-year "curr" snapshot, plotted as two synchronised 2D heatmaps (rainfall on z, temperature as a shared diverging colour scale) so the same z-axis and colour range make the two periods directly comparable.

For ETL Benchmarks:

- **Engine benchmark**: identical ingestion, SQL join, and in-memory join operations are run once on `sqlite3` + `pandas` and once on `duckdb` + `polars` , timed with `time.perf_counter` and memory-profiled with `tracemalloc` .

Further references

- ECA&D data policy and download portal: https://www.ecad.eu/download/millennium/data_policy.php
- Open-Meteo Historical Weather API docs: https://open-meteo.com/en/docs/historical-weather-api
- GeoNames export/dump directory: https://download.geonames.org/export/dump/
- `statsmodels` Unobserved Components model: https://www.statsmodels.org/stable/generated/statsmodels.tsa.statespace.structural.UnobservedComponents.html
- `statsmodels` state-space user guide: https://www.statsmodels.org/stable/statespace.html

Initialization

Parameters

```
--------------------------------------------------------------------------
ANALYSIS WINDOW:
--------------------------------------------------------------------------
- metrics:
  - Temperature TG (°C)
  - Rainfall RR (mm/day)
- months:
  - Summer [June-July] only, average values
- time-series:
  - history data: 1800-2026
  - forecast period: 2027-2036
- geographic comparison:
  - past interval: 2005-2010 average
  - current values: 2026
- geographic outlier clipping:
  - TG percentiles: 1.0-100.0
  - RR percentiles: 0.0-99.0
- geographic region clipping:
  - Europe region: lon >= -25 and lon <= 45 and lat >= 34 and lat <= 72
  - ignored regions: 80.0 km - max distance to city
  - city/station distance: 80.0 km - max distance to station
  - city by country: 250 - top n cities by country

--------------------------------------------------------------------------
```

Part 1 -- Online Dataset Retrieval

This part resolves and downloads two datasets into `temp/duckdb-demo` :

1. Climate data (TG, RR) from ECA&D, with an Open-Meteo fallback.
2. Geographic + country enrichment from GeoNames `cities500` .

Every resolved URL, downloaded size, and licence is printed for traceability, and any fallback substitution is recorded explicitly rather than silently assumed.

Data Sources References

```
Climate data source in use: 'ecad'

- GeoNames cities500
    url:     https://download.geonames.org/export/dump/cities500.zip
    licence: CC-BY-4.0 (https://www.geonames.org/export/)
    size:    12.92 MB
- GeoNames countryInfo
    url:     https://download.geonames.org/export/dump/countryInfo.txt
    licence: CC-BY-4.0 (https://www.geonames.org/export/)
    size:    0.03 MB
- ECA&D TG (mean temp)
    url:     https://knmi-ecad-assets-prd.s3.amazonaws.com/download/ECA_blend_tg.zip
    licence: ECA&D data policy: free for non-commercial research/education, acknowledge ECA&D and data providers (https://www.
ecad.eu/download/millennium/data_policy.php)
    size:    840.26 MB
- ECA&D RR (rainfall)
    url:     https://knmi-ecad-assets-prd.s3.amazonaws.com/download/ECA_blend_rr.zip
    licence: ECA&D data policy: free for non-commercial research/education, acknowledge ECA&D and data providers (https://www.
ecad.eu/download/millennium/data_policy.php)
    size:    1755.87 MB
```

Part 2 -- Local Database Loading and Benchmarks

Both pipelines perform the same three timed operations on the same source files:

1. **Ingestion** -- load the raw CSV/TSV climate and geographic files into a local database file.
2. **SQL join** -- join climate measurements to geographic + country attributes with a database-side `SELECT ... JOIN`, producing one enriched flat table.
3. **In-memory join** -- load the two source tables into DataFrames and repeat the same join using DataFrame operations only (no SQL).

`time.perf_counter` times each step; `tracemalloc` reports each step's peak memory. Results are collected into a single comparison table at the end of Part 2b.

```
EU Country Info rows: 54
```

```
Dataset ENRICHED_PL: 179,557,208 rows
```

Part 3 -- Data analysis

This part works exclusively from `ENRICHED_PL`, the `polars` DataFrame produced in Part 2, per spec. Two derived datasets are produced:

- a country-level June/July yearly-mean time series with a 10-year state-space forecast and 50%,80%,95% confidence interval
- a city-level table of "past" interval mean) vs "curr" (latest-year mean) June/July temperature and rainfall, with coordinates attached.

```
Latest year with June/July data available: 2026
```

Country time-series with forecast extrapolation

Forecast: Choice of State-Space Model Structure

Each country/variable series is one observation per year (the June/July mean), so there is no intra-series seasonal cycle to model -- seasonality was already removed by the June/July restriction and the yearly aggregation. A **local level + local linear trend** structural (`UnobservedComponents(..., level="local linear trend")`) model is used instead of, e.g., a full seasonal SARIMA:

- It is appropriate for short series (15-17 yearly points per country): it has few free parameters (level variance, trend variance, trend slope, observation noise), so it is identifiable and stable where a higher-order ARIMA would likely overfit or fail to converge.
- It naturally separates a slowly-drifting mean level from a persistent trend, which matches the expected climate-signal shape (gradual warming / rainfall drift) better than a stationary AR process.
- As a state-space model it produces a Kalman-filter-based one-step prediction and a closed-form forecast confidence interval, satisfying the requirement for a CI-bearing forecast method.

The trade-off: with this few data points per country, wide CIs are expected and appropriate -- the interval should communicate the real uncertainty of a 15-year climate signal, not be interpreted as high analytical confidence in the point forecast.

NOTE *A couple of things worth checking once this runs clean*:

With `level="local linear trend"`, both level and slope are stochastic by default (3 variance parameters to estimate via MLE), *which is a lot to ask of series with only ~15–17 points*. If you see convergence warnings or implausibly wide confidence intervals for some country/variable pairs, consider `level="local level"` (just 2 parameters) as a steadier default, or catch ConvergenceWarning explicitly alongside your existing exception handling so it's visible per-country rather than silent.

Your `len(s) < 4` guard is reasonable as a hard floor, but 4 points is still very little for a 2-parameter model plus noise variance — you may want to raise that threshold (e.g. 6–8) once real fits are happening, purely on statistical-reliability grounds rather than to avoid crashes.

Now that the loop will actually populate forecast_records, double check `CI_LEVELS` is defined earlier in the notebook (it's referenced as the default for ci_level but wasn't shown in this cell) — if it's missing, you'll get a clean `NameError` at the `def fit_forecast` line itself rather than inside the loop.

```
Forecast Countries Count: 51
European Countries Count: 54
Missing  Countries Count: 3
Missing Countries:
                   Country ISO
250  Serbia and Montenegro  CS
73           Faroe Islands  FO
138                 Monaco  MC
153                  Malta  MT
```

City data selection

```
----------------------------------------------------------------------------
OUTLIERS SELECTION:
----------------------------------------------------------------------------
- TG - Temperature percentiles:
  - PCT(1.0) = 10.6 °C
  - PCT(100.0) = 29.2 °C
- RR - Rainfall percentiles:
  - PCT(0.0) = 0 mm/day
  - PCT(99.0) = 5.6 mm/day

- Europe Cities:
  - all cities: 6,817
  - discarded cities: 948
  - kept cities: 5,869

----------------------------------------------------------------------------
```

Part 4 -- Data visualization

Data narrative

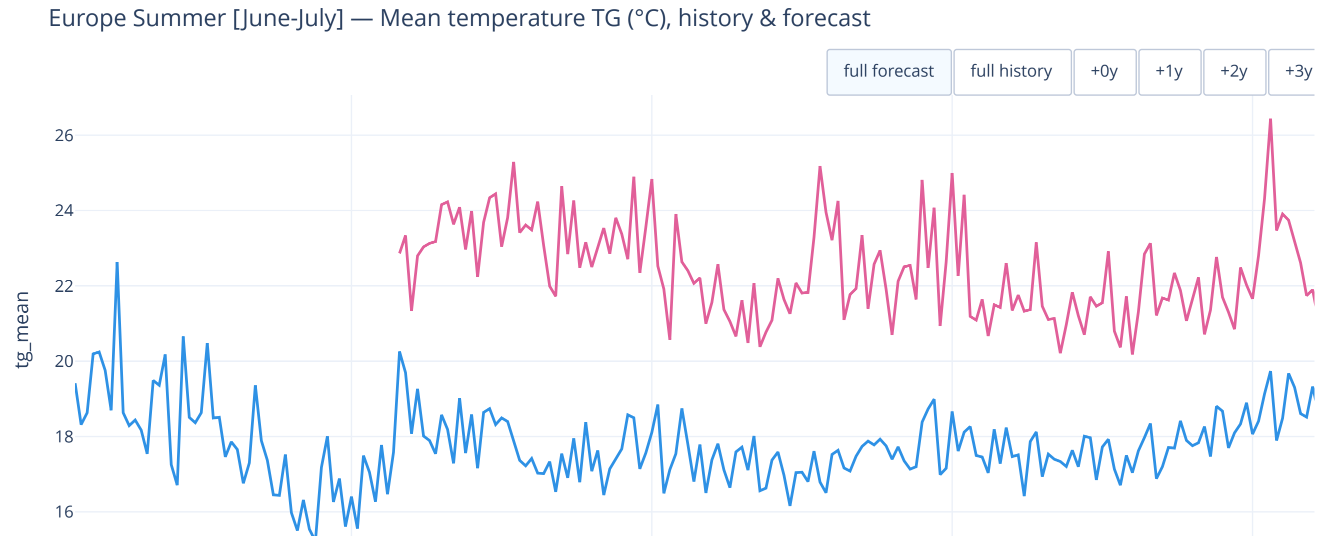
The time-series view tells the "how has each country moved" story: the historic June/July mean per year, plus a 10-year state-space extrapolation with its 95% confidence band, so the reader can see both the direction of travel and how quickly the model's certainty erodes as it looks further ahead.

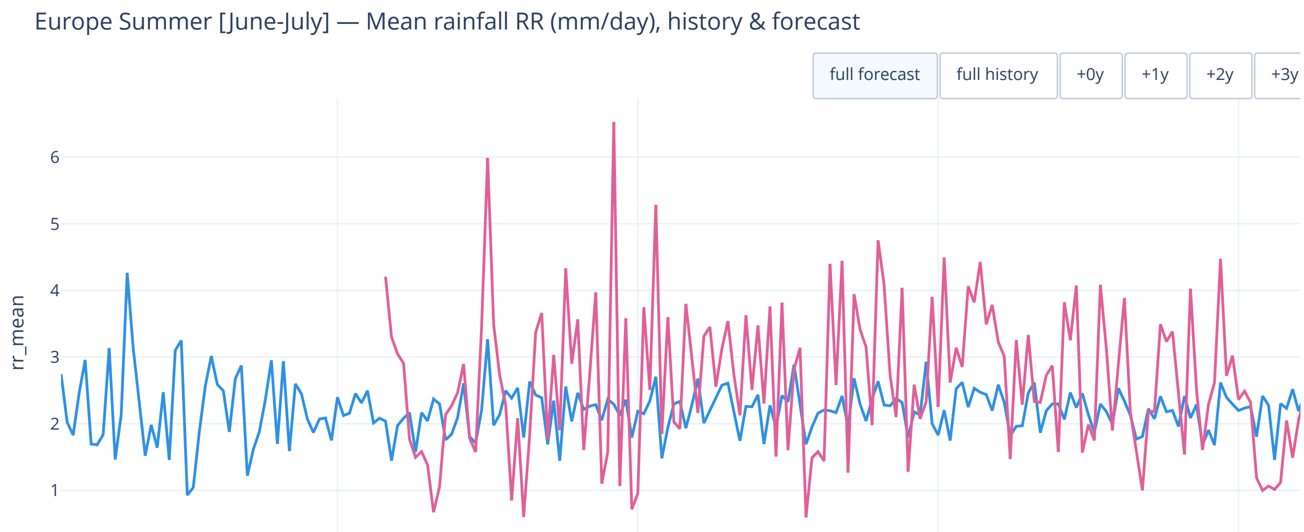
The geographic view tells the "where, right now, vs where before" story: two synchronised 2D heat-maps, with a bi-chromatic green/red for temperature/rainfall scale presents comparison between the left (2005-2010 baseline) and right (latest year) panel. In addition, a third 2d heat-map shows the difference between time intervals.

The data distribution plots shows city pairs (TG,RR), colored on latitude scale, with outline values separation.

The measurement 3d plots shows differences (curr-past) over a geographic map, with reversed z-axis mapped to rainfall and color to temperature. Points are connected to base plan with a blue line if rainfall increased or green line if rainfall decreased.

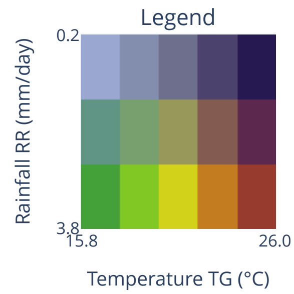
Part 4.a - Time-Series (History and Forecast)





Part 4.b - Geographic Visualization

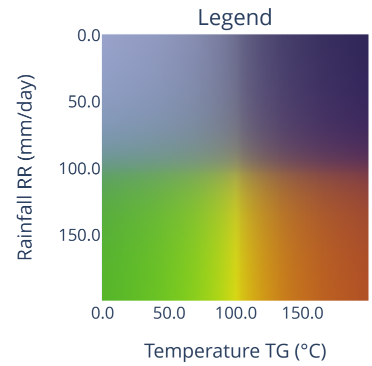
City temperature/rainfall — bivariate heatmap



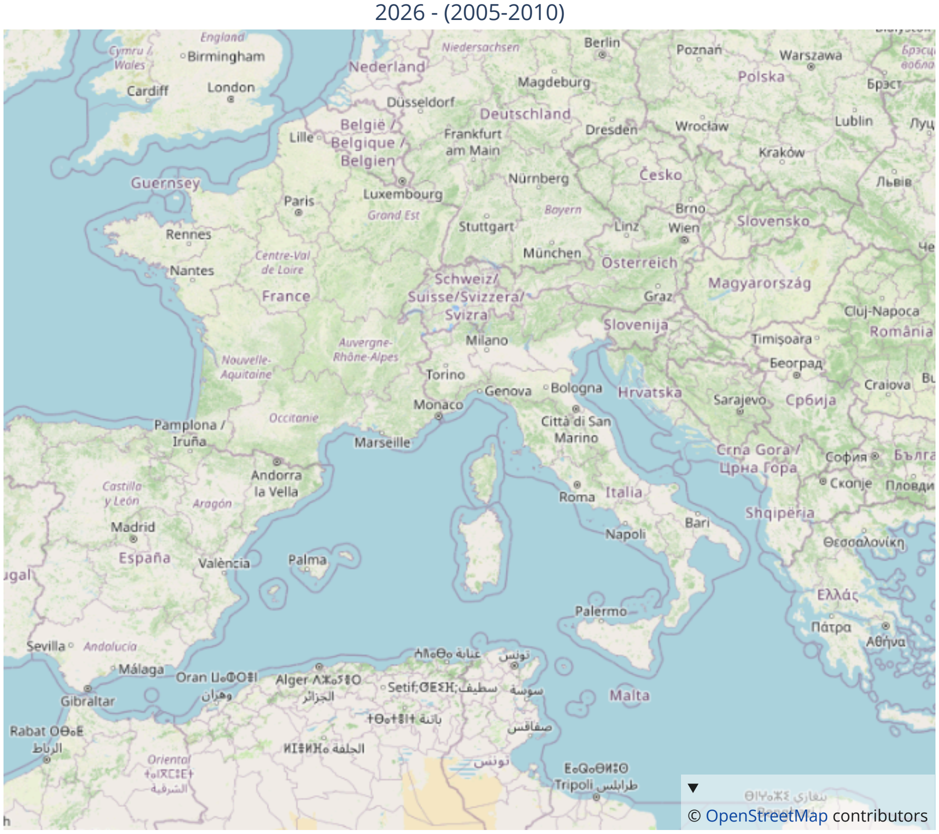
Past (2005-2010)

Current (2026)

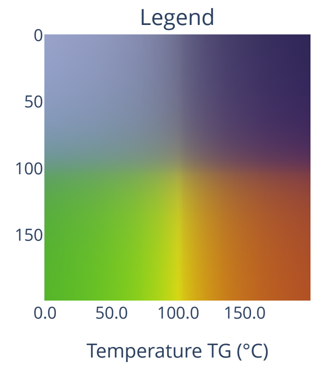
City temperature/rainfall differences — bivariate heatmap



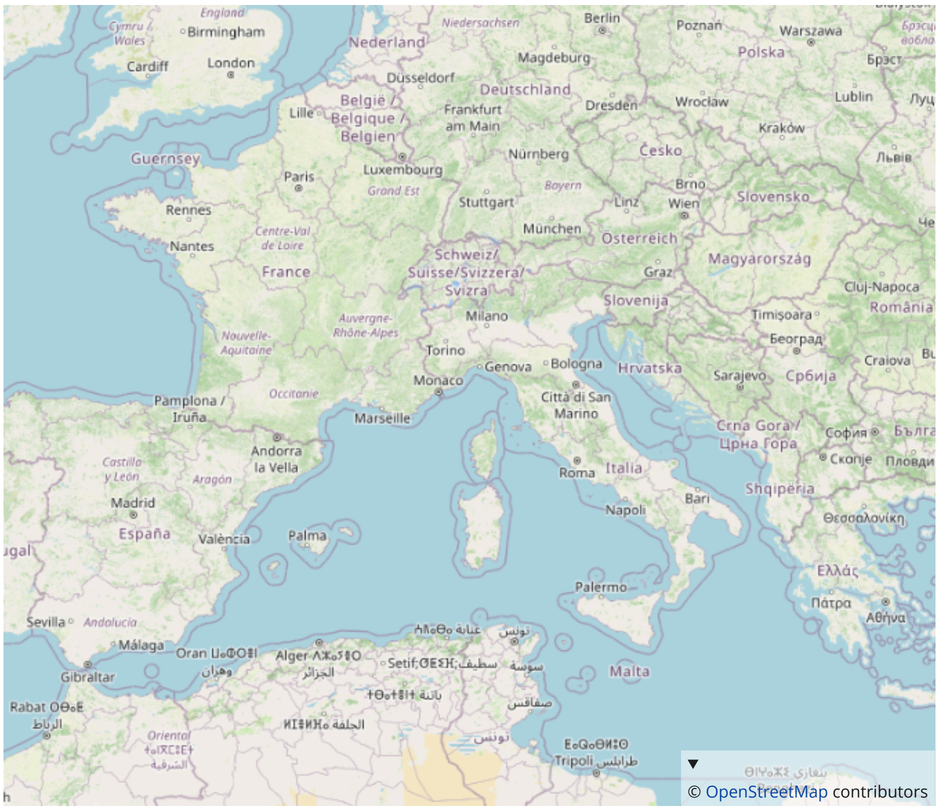
2026 - (2005-2010)



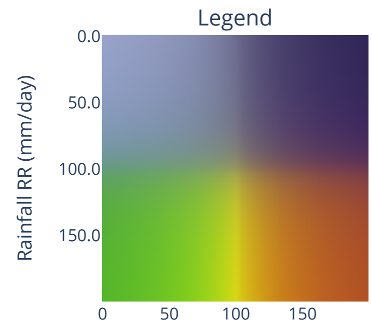
City temperature differences — univariate heatmap



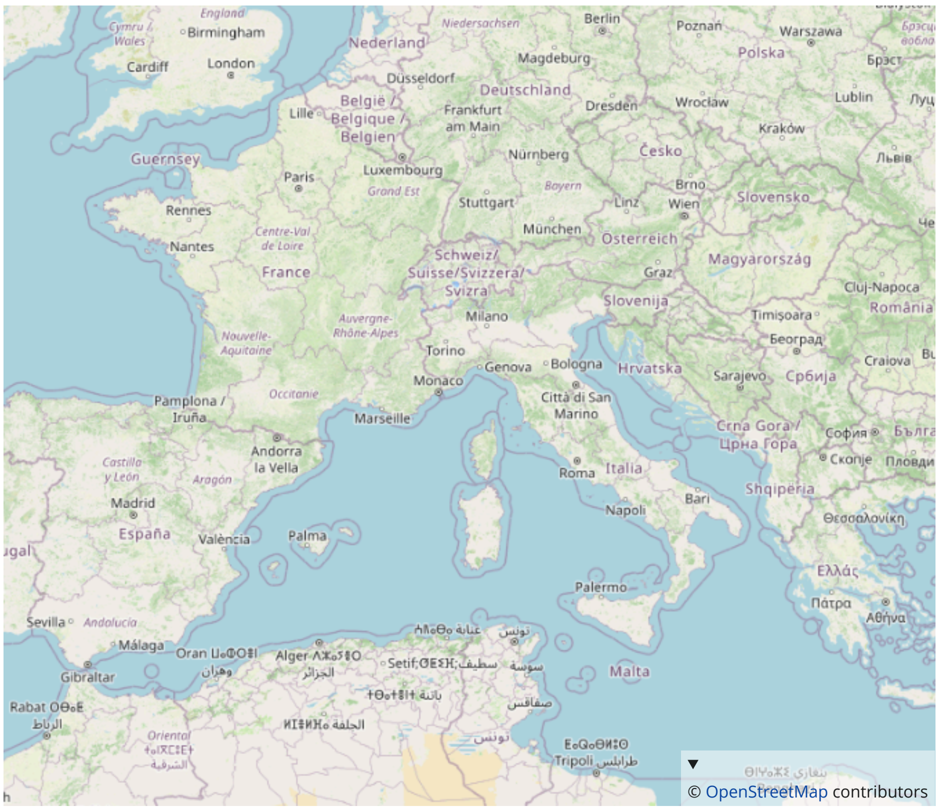
2026 - (2005-2010)



City rainfall differences — univariate heatmap

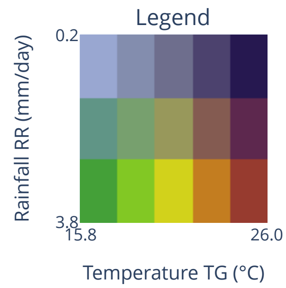


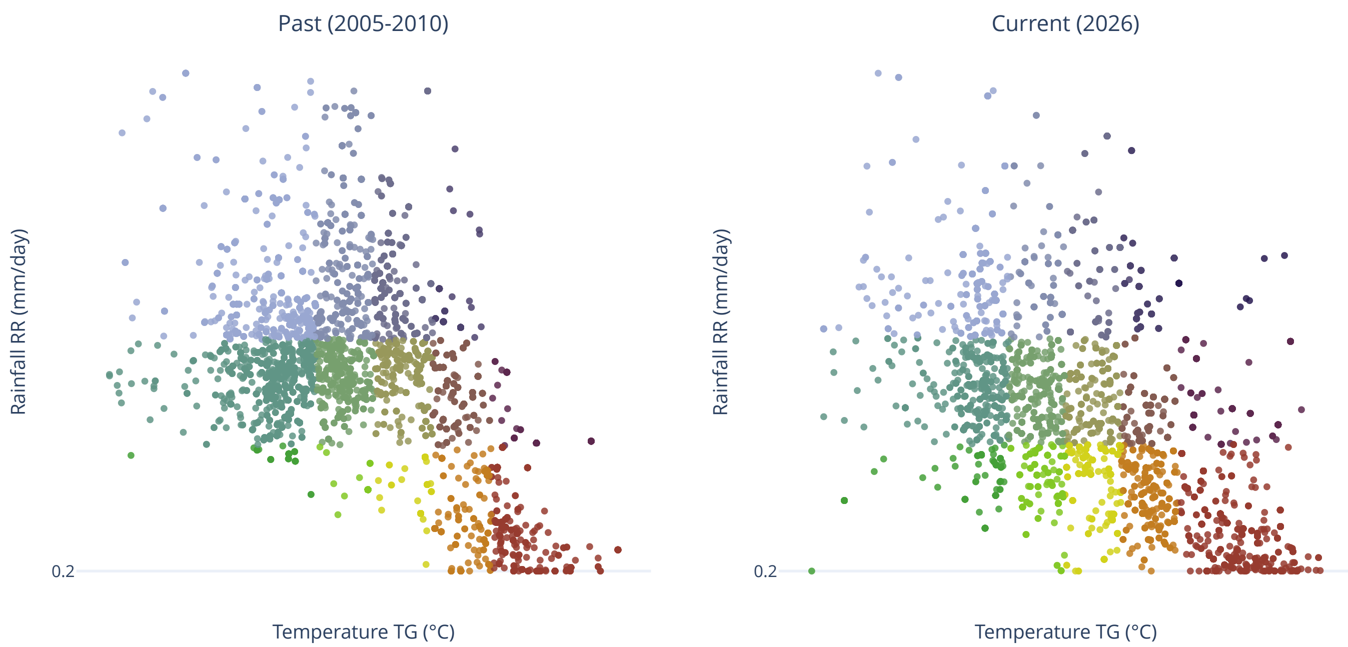
2026 - (2005-2010)

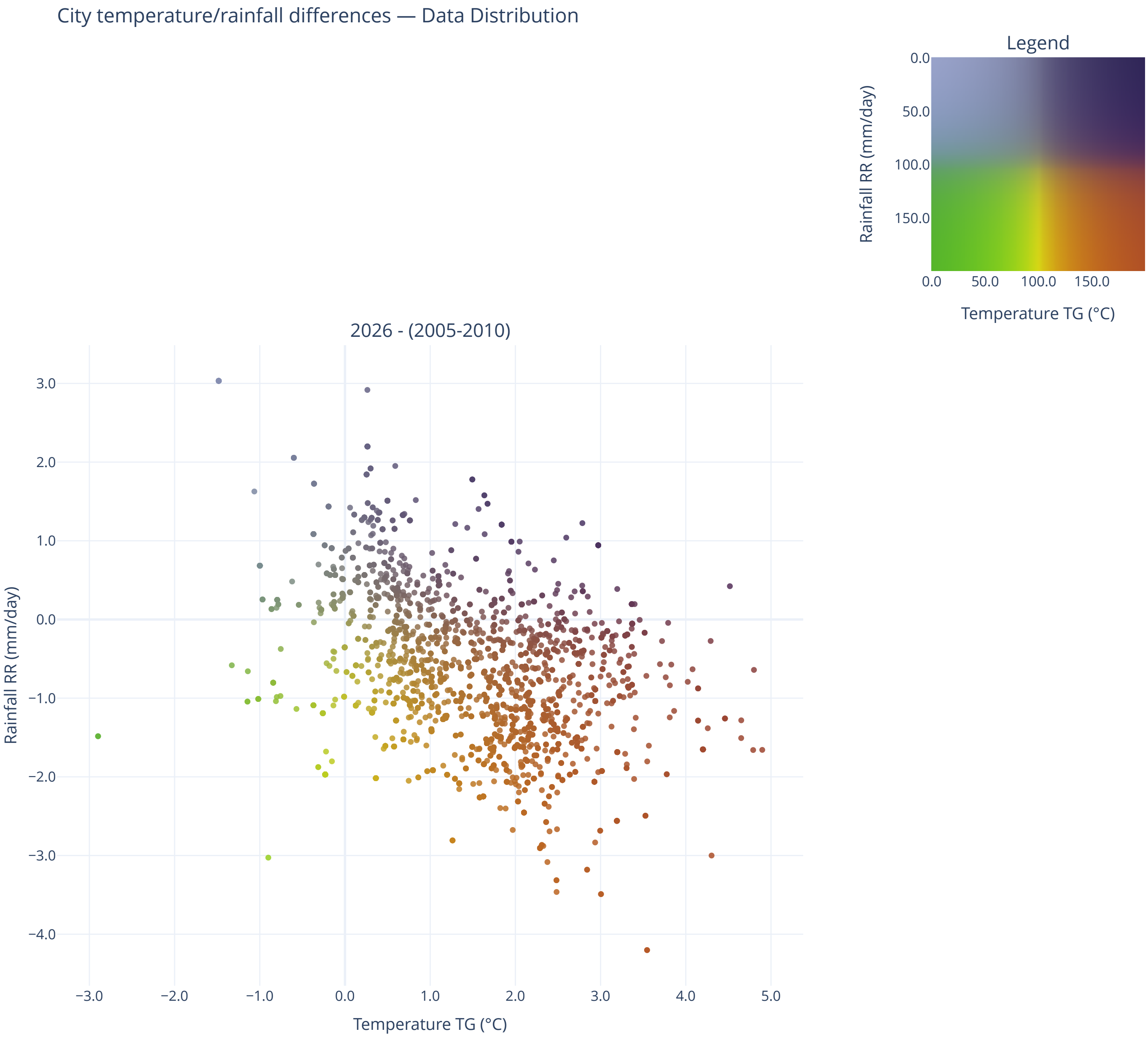


Part 4.c - Data Distribution and Outliers

City temperature/rainfall — Data Distribution

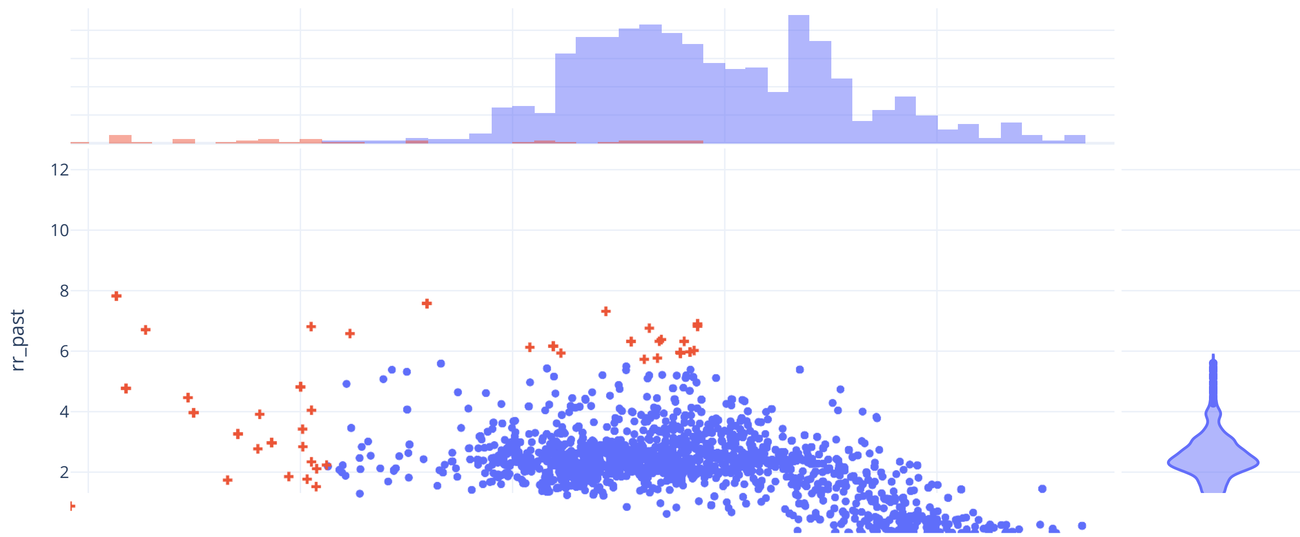




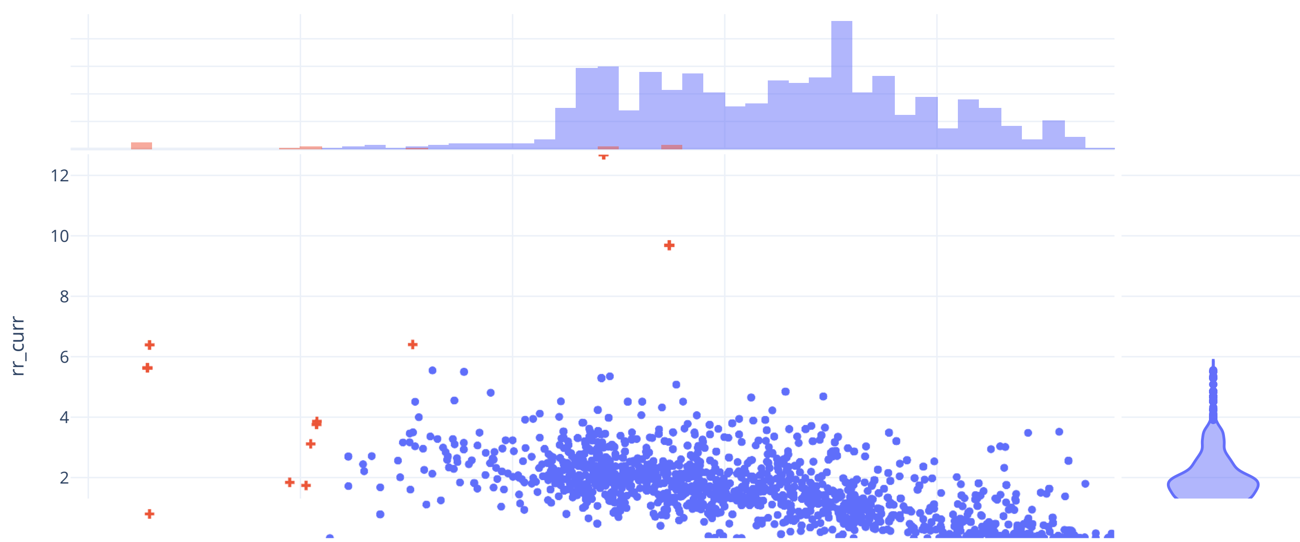


Part 4.d - Outlier Selection

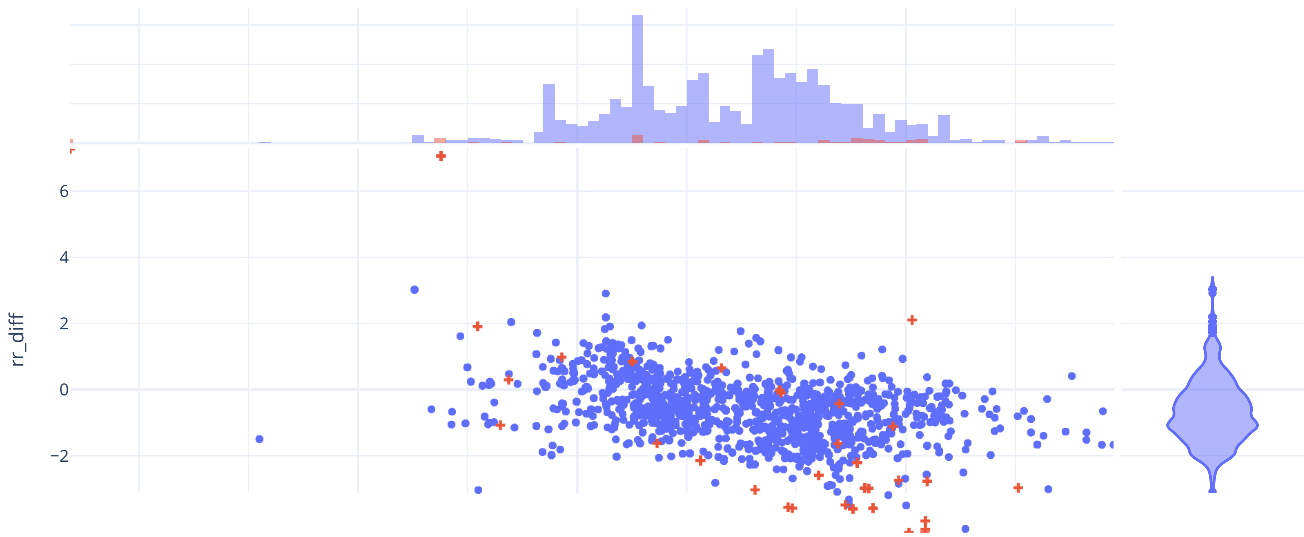
City: temperature, rainfall -- Past baseline (2005-2010)



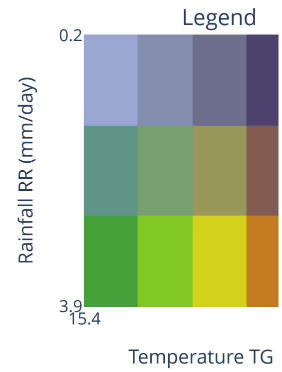
City: temperature, rainfall -- Current (2026)



City: temperature, rainfall -- (2026 - 2005:2010) Differences



City temperature/rainfall Summer [June-July] — bivariate comparison



Past (2005-2015)

Current (202

© OpenStreetMap contributors

Part 4.e - Measurements for Cities with Stations

City temperature (height) / rainfall (colour) -- current vs past difference
2026 - (2005-2010)